RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Saturday, 12 September 2026, 2:08 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:09 PM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Recursive Divide and Conquer function
int findPair(int arr[], int low, int high, int x, int *res1, int *res2) {
    if (low >= high) return 0; // Base case: range exhausted, no pair found

    int current_sum = arr[low] + arr[high];

    if (current_sum == x) {
        *res1 = arr[low];
        *res2 = arr[high];
        return 1; // Pair found
    }

    // Divide step based on the sum comparison
    if (current_sum < x) {
        return findPair(arr, low + 1, high, x, res1, res2); // Shrink from left
    } else {
        return findPair(arr, low, high - 1, x, res1, res2); // Shrink from right
    }
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    if (scanf("%d", &x) != 1) return 0;

    int res1, res2;
    if (findPair(arr, 0, n - 1, x, &res1, &res2)) {
        printf("%d\n%d\n", res1, res2);
    } else {
        printf("No\n");
    }

    return 0;
}

```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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